A 9.1-ENOB 6-mW 10-Bit 500-MS/s Pipelined-SAR ADC With Current-Mode Residue Processing in 28-nm CMOS

Kyoung-Jun Moon, Dong-Shin Jo, Wan Kim, Michael Choi, Hyung-Jong Ko, and Seung-Tak Ryu, *Senior Member, IEEE*

***Abstract*— This paper introduces a current-mode residue processing technique in a pipelined-successive-approximation register (SAR) analog-to-digital converter (ADC), which extends the operation speed of a single-channel ADC utilizing low-impedance-based signaling. A 10-bit pipelined-SAR ADC with featured building blocks such as a degenerated gm-cell as an open-loop residue amplifier, a switched-current mirror for sample-and-hold (S/H) function, and a split current digital-to-analog converter (DAC) for current-domain SAR conversion achieves a 500-MS/s conversion-rate under a 1.0-V supply. With background inter-stage mismatch calibration, a prototype ADC fabricated in a 28-nm CMOS process achieves 56.6-dB signal-to-noise-and-distortion ratio (SNDR) at a Nyquist input, resulting in a Walden figure of merit (FoM) of a 21.7-fJ/conversion-step.**

***Index Terms*— Analog-to-digital converter (ADC), current-domain, current-mode residue processing, gm-cell, open-loop amplifier, pipelined-successive-approximation register (SAR), split current digital-to-analog converter (DAC).**

I. Introduction

RECENTLY reported pipelined-successive-approximation register (SAR) analog-to-digital converters (ADCs) combining moderate resolution sub-stage SAR ADCs with a residue amplifier have shown not only excellent power efficiency but also high conversion rates [1]–[10]. Although the conversion speed of ADCs with a pipelined-SAR architecture can be significantly enhanced with time-interleaving (TI) techniques [3]–[8], mismatches between sub-channels in TI ADCs present a considerable design bottleneck. With the channel mismatch problems in highly interleaved ADCs as motivation, some previous designs have been developed to enhance the single-channel speed of pipelined-SAR ADCs [9], [10]. Various architectural modifications of ADCs in their internal SAR ADCs have been proposed, such as multi-bit/cycle conversion [9] and a loop-unrolled structure [10]. Nevertheless, the residue amplifier required for pipelining has always been a major design challenge for speed, gain accuracy, and power consumption.

Design techniques that replace opamp-based residue amplifiers with simple open-loop amplifiers have been exploited for low power and high speed. As reported in [4]–[6] and [9], the integrated dynamic amplifiers can provide a very power-efficient solution, but the voltage-to-time conversion process makes the structure very sensitive to process–voltage–temperature (PVT) variation. The background gain calibration [6] and the replica-aided dynamic amplifier [9] schemes could enhance the robustness, but they accompany design complexity and additional power consumption. In addition, the dynamic amplifiers basically require an integration time period or high supply voltage for high gain and a high signal-to-noise ratio (SNR) [11], and this would limit the speed of the residue generation. This time and supply dependency problem of the dynamic amplifiers could be alleviated with a traditional open-loop voltage amplifier with a resistive load [10]. However, the *RC* time constant with the capacitive load of the subsequent SAR ADC can also limit the operation speed.

For a power-efficient high-speed single-channel pipelined-SAR ADC, a current-mode open-loop residue processing scheme [12] is introduced in this paper. Even though current-mode pipelined ADCs have been previously reported as [13]–[15], all of them utilized a closed-loop approach. The proposed ADC employs a gm-cell as an open-loop residue amplifier, and a high-speed current-steering digital-to-analog converter (DAC) is utilized for the fine SAR ADC for low-impedance-based high-speed signaling.

This paper is organized as follows. In Section II, the proposed current-mode open-loop residue processing scheme is discussed. The detailed circuit implementation and an analysis of the proposed pipelined-SAR ADC are then provided in Section III. The measurement results of the prototype ADC are presented in Section IV, followed by conclusions in Section V.

II. Proposed Current-Mode Open-Loop Residue Processing

A conventional pipelined-SAR ADC with a voltage-mode open-loop residue processing scheme is modeled in Fig. 1. It consists of an open-loop residue amplifier with a resistive load (R_L) and an N-bit coarse and M-bit fine SAR ADCs with a capacitive DAC (C-DAC). The output voltage (V_{OUT}) of the residue amplifier is obtained as

$$V_{\mathrm{OUT}}(t) = V_{\mathrm{RES}} \cdot G_{\mathrm{m}} \cdot R_L \cdot (1 - e^{-\frac{t}{R_L \cdot C_L}}). \quad (1)$$

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K.-J. Moon and S.-T. Ryu are with the Department of Electrical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, South Korea (e-mail: vandoren@kaist.ac.kr; stryu@kaist.ac.kr).

D.-S. Jo is with SK Hynix, Icheon 17336, South Korea.

W. Kim, M. Choi, and H.-J. Ko are with Samsung Electronics, Hwaseong 18448, South Korea.

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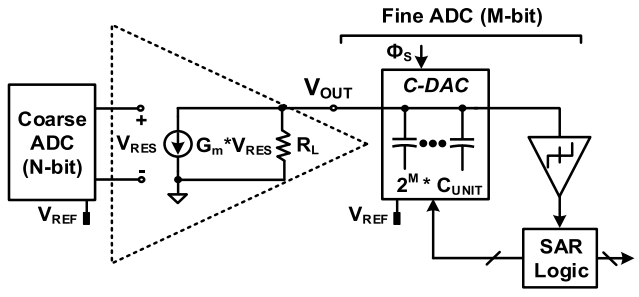


Fig. 1. Voltage-mode residue processing in pipelined-SAR ADC.

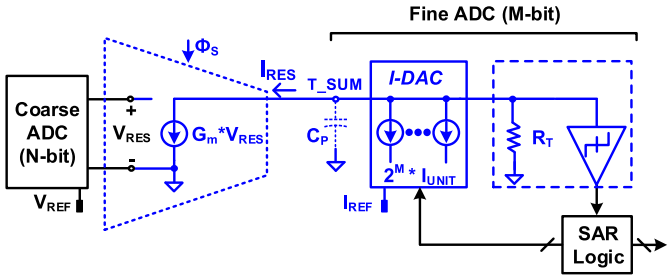


Fig. 2. Proposed current-mode residue processing in pipelined-SAR ADC.

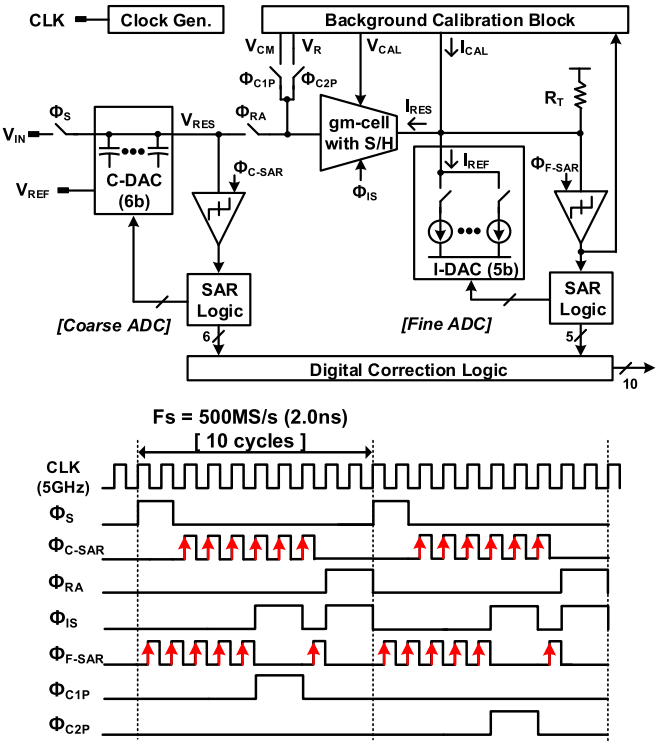


Fig. 3. Block and timing diagram of the 10-bit 500-MS/s pipelined-SAR ADC.

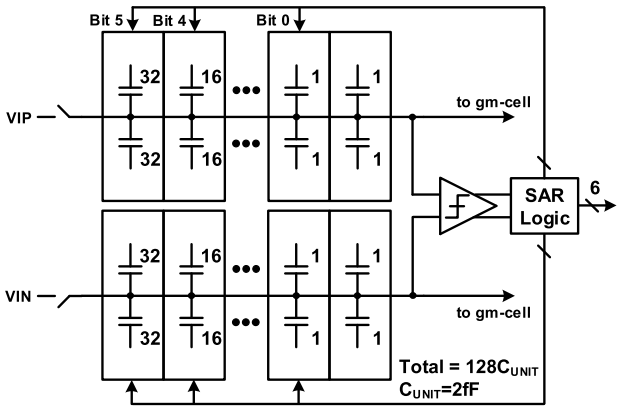


Fig. 4. Implementation of the 6-bit coarse ADC.

With a residue voltage (V_{RES}) generated from the coarse SAR ADC as the input of the amplifier, the operation speed of the single-pole open-loop amplifier is determined by the RC time constant (τ) at the output node of the amplifier. In particular, the gain of the residue amplifier ($\propto R_L$) and the total capacitance ($2^{\mathrm{M}} \cdot C_{\mathrm{UNIT}}$) of the C-DAC in the fine SAR ADC are the parameters that provide the largest contribution to the RC time constant.

In accordance with trends of recently advanced CMOS processes, the C-DAC in the fine SAR ADC can be implemented with a small C_{UNIT} of less than 1 fF, showing excellent matching performance [11]. Also, it is possible to minimize the load resistance (R_L) by reducing the residue gain of the amplifier, which further alleviates the non-linearity characteristic of the residue amplifier with a small output swing level. The total capacitance ($2^{\mathrm{M}} \cdot C_{\mathrm{UNIT}}$) of the C-DAC, however, directly affects the performance of the fine ADC, such as kT/C noise, comparator kickback noise, and signal attenuation due to the input parasitic capacitance of the comparator. Therefore, as the resolution of the fine ADC increases, a larger total capacitance is required for the C-DAC to realize noise-immune pipelining. In addition, the influence on the RC time constant by additional parasitic capacitances such as a junction and metal line capacitance at the output node of the amplifier is not negligible. Hence, voltage-mode open-loop residue processing with a C-DAC for fine SAR restricts the RC time constant minimization for high-speed operation.

By replacing the conventional C-DAC with a current-steering DAC (I-DAC), as depicted in Fig. 2, the proposed current-mode open-loop residue processing scheme alleviates the capacitive load, and thus, facilitates higher speed residue generation. The proposed pipelined-SAR ADC consists of a gm-cell with a sample-and-hold (S/H) function, an I-DAC with a reference current (I_{REF}), and a comparator with a trans-impedance resistor (R_{T}) for fine SAR conversion. The gm-cell converts the residue voltage (V_{RES}) to current (I_{RES}) and holds it by utilizing a sampling clock (Φ_{S}). The settling time required to amplify the residue through the gm-cell, similar to the voltage-mode pipelined-SAR ADC as (1), is decided by the parasitic capacitance (C_{P}) and load resistance (R_{T}) at the current summing node (T_SUM). Unlike the total capacitance (C_L) of the C-DAC in Fig. 1, the parasitic capacitance (C_{P}) is independent of the signal level ($\propto R_{\mathrm{T}}$) at the node of T_SUM determining the SNR of the fine ADC. By minimizing the C_{P}, a low RC time constant can, therefore, be realized without sacrificing the SNR performance. As the voltage swing level at the node of T_SUM becomes smaller, the I-DAC has more voltage headroom, and thus, the linearity and noise performance

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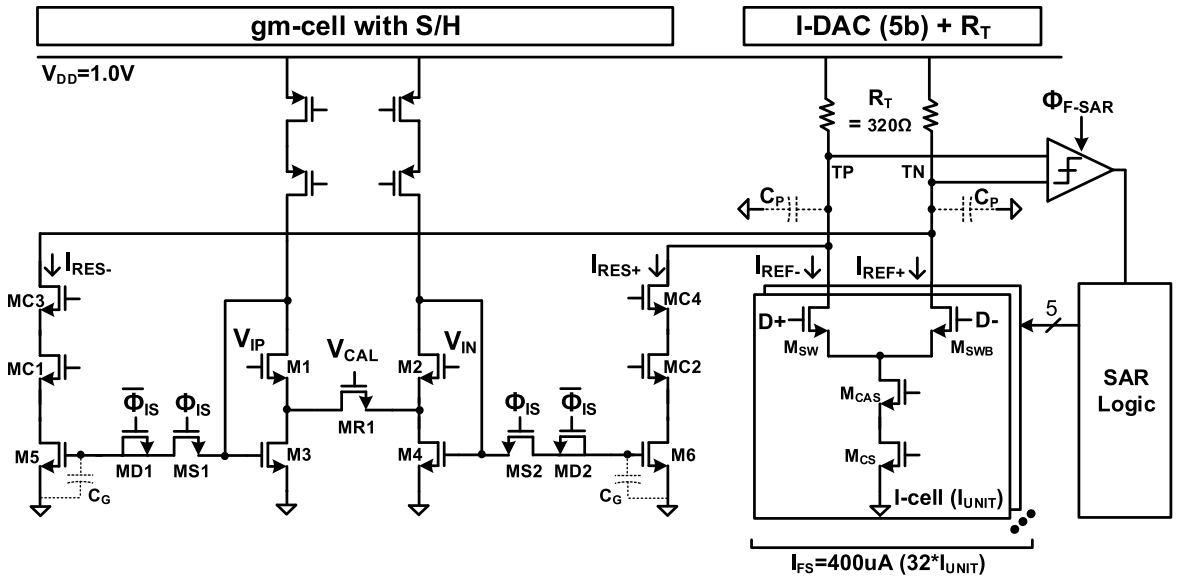


Fig. 5. Circuit Schematic of the gm-cell with S/H and the 5-bit fine SAR ADC.

of the fine ADC can be secured. Hence, the current-mode residue processing technique is more suitable for low supply. The parasitic capacitance (C_P) consists of the device junction capacitance and metal line parasitic of the I-DAC, and a very small value can be obtained by proper biasing in the unit current cell and a compact layout. The estimated C_P of the prototype ADC is below 30 fF. The proposed current-mode residue processing scheme results in low-impedance signaling for fast residue settling while realizing a noise-optimized design.

III. Circuit Implementation

A. Overall Architecture and Coarse ADC

Fig. 3 presents an overall block diagram of the proposed ADC with its operational timing diagram. The proposed ADC is composed of a 6-bit voltage-mode coarse SAR ADC, a gm-cell with current-mode S/H function for residue amplification, and a 5-bit fine SAR ADC with a current-steering DAC (I-DAC). With a 1-bit redundancy between the coarse and fine SAR ADCs, the entire ADC generates a total of 10-bit digital output. A background calibration function is embedded on a chip to calibrate the errors between the coarse and fine stages. The 10-bit A/D conversion is completed within ten internal clock cycles. The input voltage (V_{IN}) is sampled on the C-DAC for 1.5 clock cycles and 6-bit coarse A/D conversion is then conducted for the following 6.5 clock cycles. After the coarse A/D conversion is completed in a total of eight cycles, the residue voltage (V_{RES}) remaining on the C-DAC is converted into current (I_{RES}) by the gm-cell for the last two clock cycles. The residue current is sampled and held in the gm-cell by the S/H function at the end of the tenth clock cycle. During the first five clock cycles in which the coarse ADC samples the next input and conducts the SAR operation with it, the 5-bit fine ADC with its I-DAC conducts SAR operation with I_{RES}. While the coarse A/D conversion is still in process, during the three clock cycles after the fine SAR operation is completed, the gm-cell and the fine ADC conduct offset and gain calibration with the relevant calibration circuitry.

As shown in Fig. 4, a split binary-weighted a C-DAC with a top-plate sampling scheme is employed in the voltage-mode coarse SAR ADC [16]. This switching scheme enables high-speed DAC switching with simple SAR logic and maintains the common-mode voltage of the gm-cell constant regardless of the code. The total C-DAC capacitance of the 6-bit coarse SAR ADC is designed to be 256 fF considering the kT/C noise and capacitor mismatch for a 10-bit resolution with a 1.6-V full-scale differential swing. A total error of about 25 mV of the comparator offset and the DAC settling in the coarse ADC can be tolerated by the 1-bit inter-stage redundancy.

B. Current-Mode Back-End

Fig. 5 shows a detailed block diagram of the residue-amplifying gm-cell with S/H and the 5-bit fine SAR ADC. The gm-cell with S/H function is composed of a flipped voltage follower (FVF)-structured input stage (M1–M4) [17] with a degeneration resistor (MR1) for linear voltage-to-current conversion and a switched-current mirror (V–I) to hold the charges on its gate capacitance (C_G) for current sampling (M5, M6, MS1, MS2). For linearity of the voltage-to-current conversion gain, a triple cascode current mirror (M5, M6, MC1–MC4) and dummy switches (MD1, MD2) are utilized. The fine ADC consists of a 5-bit I-DAC, a voltage comparator with a trans-impedance resistor (R_T) and a SAR logic. The differential residue voltage of V_{IP} and V_{IN} are converted into currents, I_{RES-} and I_{RES+}, by the gm-cell. For the SAR operation, the residue current sampled by the clock Φ_{IS} is compared with the currents from the 5-bit I-DAC, I_{REF+}, and I_{REF-}. All current sources in the gm-cell and the I-DAC

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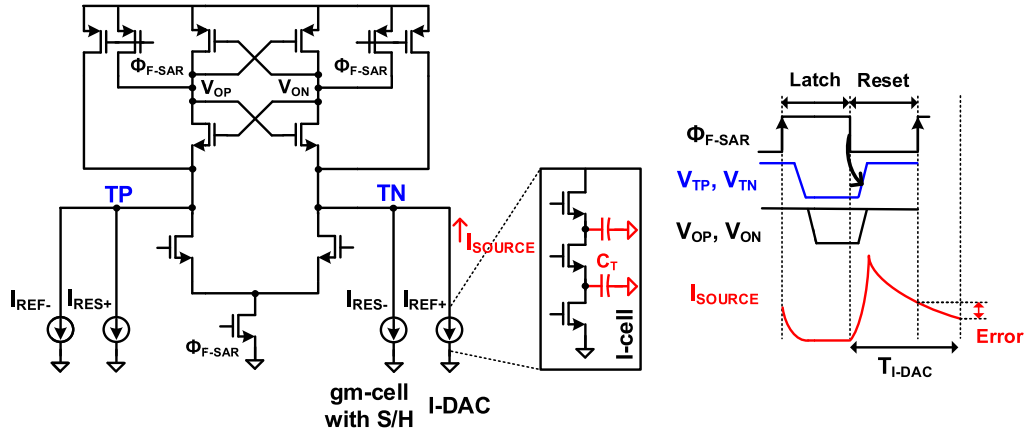


Fig. 6. SAR operation with current comparison.

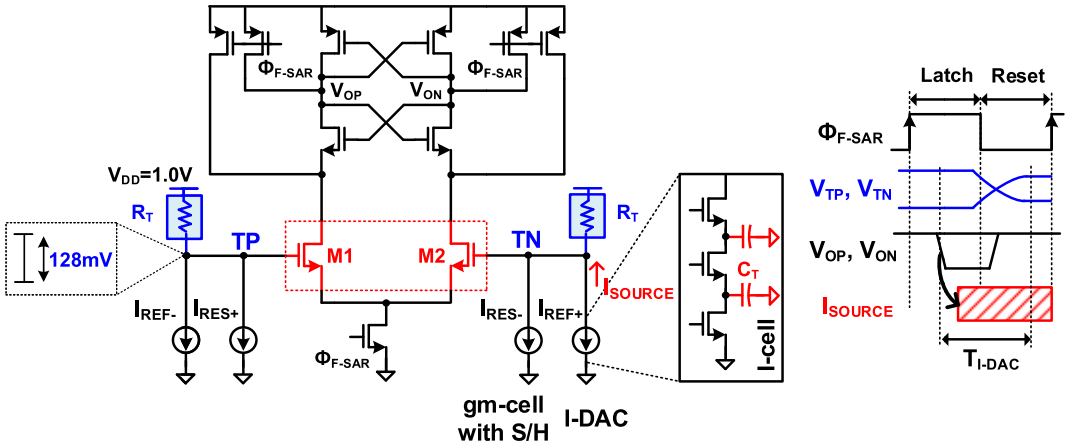


Fig. 7. SAR operation with voltage comparison with R_T.

employ only an NMOS transistor for a supply voltage under 1.0 V, enhancing the operation speed and mitigating design complexity.

1) Voltage Comparator With R_T: Since both the amplified residue and the DAC signal in the fine ADC are in current mode, it would be natural to choose a current comparator that takes the current signals directly as its input, as conceptually shown in Fig. 6. However, with this structure, a speed issue arises when the reset time of the comparator becomes short (i.e., in high speed operation) due to the slow DAC turn-on time: At the end of the latching phase, both V_{TP} and V_{TN} reach ground level, and thus, all current sources enter the triode region. During the following reset phase, V_{TP} and V_{TN} are reset to the supply level, and the current sources turn on for DAC operation by initially charging the parasitic capacitance (C_T) of the internal nodes. As the reset time becomes shorter, however, the current sources cannot be fully turned on due to the insufficient parasitic charging. This results in insufficient DAC settling, which leads to erroneous SAR conversion. In order to avoid this problem in the gm-cell and the I-DAC, a small resistive load (R_T) is connected to the current output, as shown in Fig. 7. The R_T converts the current output into voltage, and thus, a voltage comparator conducts the comparison. Therefore, the current sources are always turned on regardless of the reset operation of the comparator and the DAC settling time for the SAR operation can be ensured, which enables high-speed SAR operation of the fine ADC.

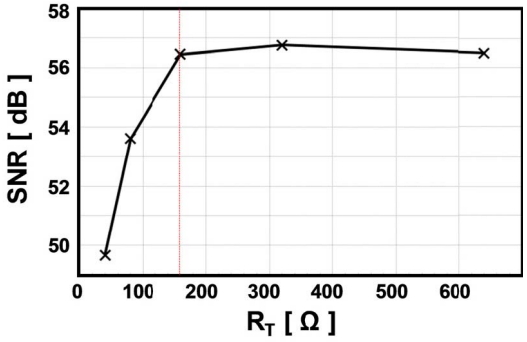


Fig. 8. Transient noise simulation result of ADC's SNR versus R_T.

Accordingly, the fine SAR ADC is constructed with a current DAC that drives the R_T and a strong-arm comparator. In terms of residue settling speed, a smaller R_T is preferred, and therefore, R_T is chosen to be as small as possible until the voltage swing level at the summing node (TP, TN) limits the SNR of the fine ADC. In this structure, the noise

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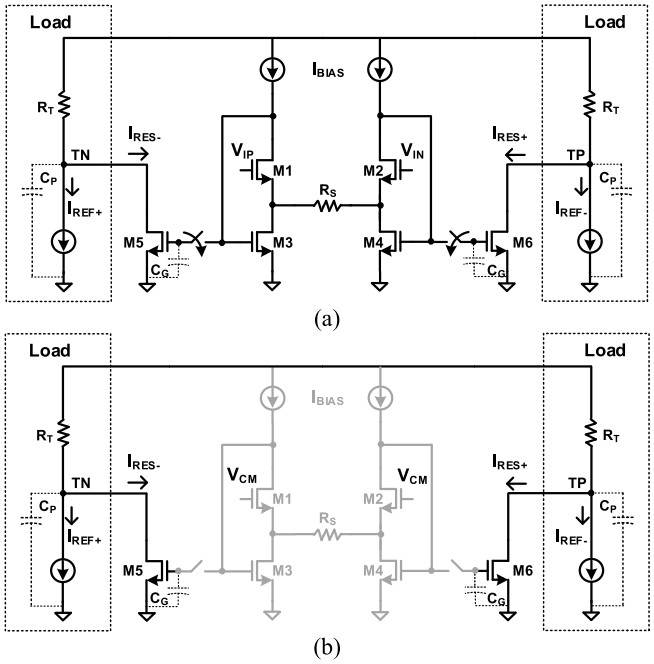


Fig. 9. Circuit schematic of the gm-cell (a) in the sampling phase and (b) in the holding phase.

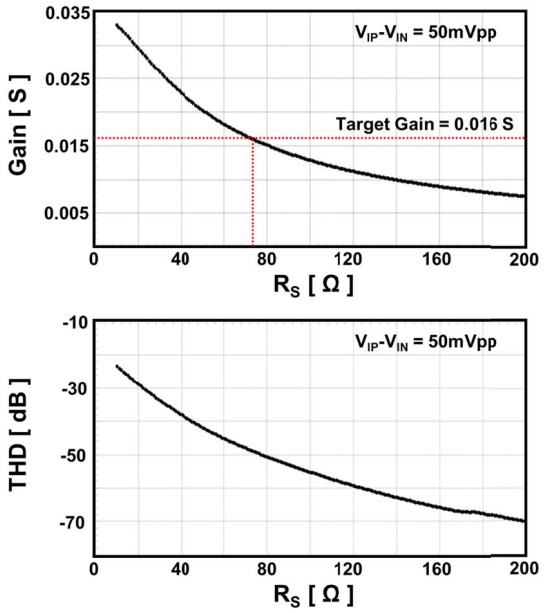


Fig. 10. Simulated gain and THD versus R_S in the gm-cell.

characteristics of the fine ADC are determined mainly by the thermal noise of the comparator. A smaller R_T at the node of TP and TN leads to a higher common-mode level at the input of the comparator, which causes a faster but noisy operation of the comparator with a normal threshold transistor. The effective overdrive voltage of the comparator input transistor (M1, M2) is, thus, reduced by utilizing a high threshold device to minimize the noise influence of the comparator.

Fig. 8 presents the transient noise simulation results showing the SNR performance of the overall 10-bit ADC according to the R_T value with the full-scale current of 400 μA of the I-DAC. The simulated minimum value of R_T at which the noise impact of the comparator in Fig. 7 does not degrade the overall ADC's SNR of about 57 dB (dominated by the noise of the quantization and the gm-cell) is roughly 160 Ω. Considering the process and temperature variations of the resistance (>20%), an R_T value of 320 Ω is, thus, utilized and the residue swing range at the summing node (TP, TN) is around 128 mV.

2) gm-Cell With S/H: Fig. 9 depicts the operational principle of the gm-cell in (a) the sampling phase and (b) the holding phase. In the sampling phase, the small-signal voltage-to-current conversion gain of the gm-cell is

$$\frac{I_{RES+} - I_{RES-}}{V_{IP} - V_{IN}} \approx \frac{1}{\frac{1}{g_{m1} \cdot g_{m3} \cdot r_{o1}} + \frac{R_S}{2}} \tag{2}$$

where g_{m1} and g_{m3} are the trans-conductance of M1 and M3, respectively, and r_{o1} is the drain–source resistance of M1. The simulation results of the gm-cell gain and the total harmonic distortion (THD) according to the degeneration resistor value, R_S, are shown in Fig. 10. With R_S and the negative feedback characteristic of the FVF, the gm-cell obtains a well-defined voltage-to-current gain while securing linearity. In this design, the maximum input voltage of the gm-cell as the residue from the coarse ADC is about 50 mV. With the R_T value of 320 Ω and voltage swing at the output (TP, TN) of around 128 mV, as mentioned above, the full-scale output current of the gm-cell (I_{RES+}, I_{RES-}) is differential 800 μA. The gm-cell with the target voltage-to-current gain of 0.016 S should satisfy the 5-bit linearity. Hence, R_S is designed to be about 80 Ω with a −47-dB THD. With a THD drop of less than 1 dB, an NMOS transistor in the triode region, the MR1 shown in Fig. 5, realizes R_S for gain calibration purpose (PVT-insensitivity of the open-loop amplifier) with a controlled gate voltage of V_{CAL}. While the gm-cell holds the sampled residue current owing to the gate capacitance (C_G) of the mirror transistor, as shown in Fig. 9(b), the gate capacitance of the input transistor of the FVF stage should be reset to remove the memory effect. Although the input nodes of the gm-cell can be reset to the ground level for power savings during the holding phase, as the recovery time from reset to the subsequent sampling phase can degrade the settling time, a V_{CM} of around 0.55 V is utilized for input reset. This reset operation is also required for the calibration mode, as will be discussed in detail in Section III-D.

3) Split-Current Source DAC: The current DAC structure and its switching algorithm have been modified in this work to improve the DAC settling speed of the fine SAR ADC. A representative conventional switching technique of a 5-bit binary-weighted current-steering DAC is shown in Fig. 11(a). For MSB conversion, a differential zero signal (IREF_diff = 0) is generated by connecting the MSB current source of $16I_{UNIT}$ to one output (e.g., I_{REF-}) while the remaining current sources, whose total current is also $16I_{UNIT}$, are connected to the other output (I_{REF+}). When the MSB is determined to be one by the SAR operation, the MSB-1 current source of $8I_{UNIT}$ is connected to the negative output, and the $16I_{UNIT}$ is switched back to the positive output for the following MSB-1 conversion.

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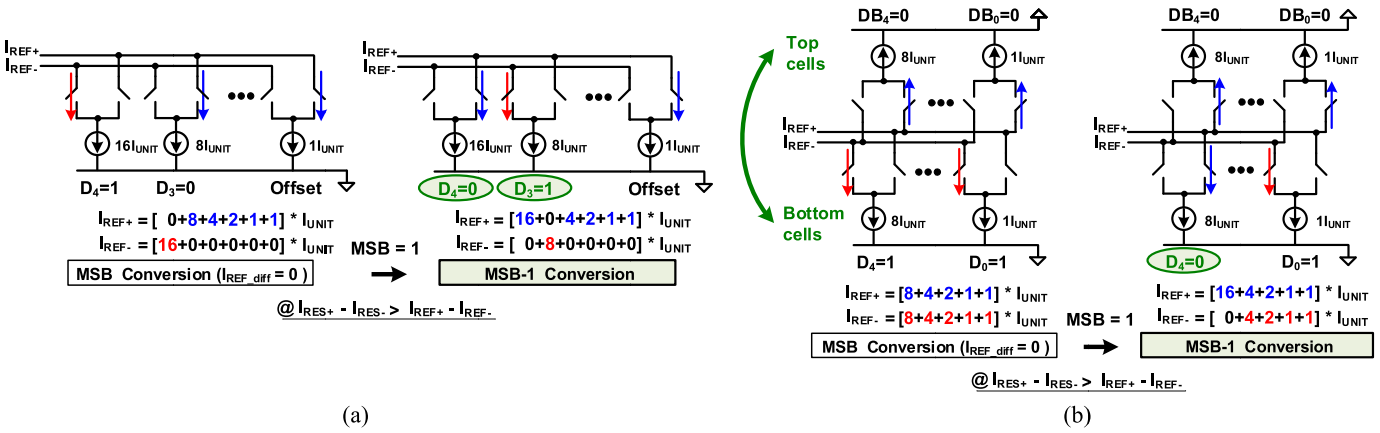


Fig. 11. Switching algorithm of the (a) conventional current-steering DAC and (b) proposed split-current source DAC.

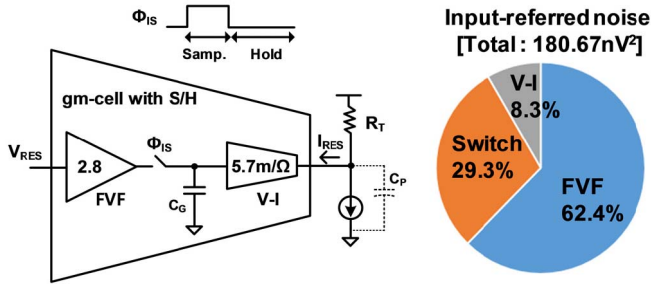


Fig. 12. Noise analysis of the gm-cell with S/H.

TABLE I
SIMULATED NOISE CONTRIBUTION

	Noise [nV²]	[%]
Quantization	203.45	37.7
Coarse ADC	101.1	18.7
gm-cell with S/H	180.67	33.5
I-DAC + R_T	16.51	3.0
Fine Comparator	38.15	7.1
Total	539.88	100
Estimated SNR	57.73 dB	ADC Input 1.6V_{PP}

This switchback operation increases the control logic delay, which degrades the conversion speed, as well as the power efficiency. Fig. 11(b) shows the proposed split current source DAC that eliminates the switchback operation in the two-level referenced DAC. Each binary-weighted current source of the conventional DAC is split in half, with top and bottom cells, similar to the split capacitor DAC in [16]. For MSB conversion, a differential zero is generated by connecting all the bottom cells to the negative output, while all the top cells are connected to the positive output. When the MSB is determined to be one, only the $8I_{UNIT}$ in the bottom cells switches to the positive output for the next reference generation.

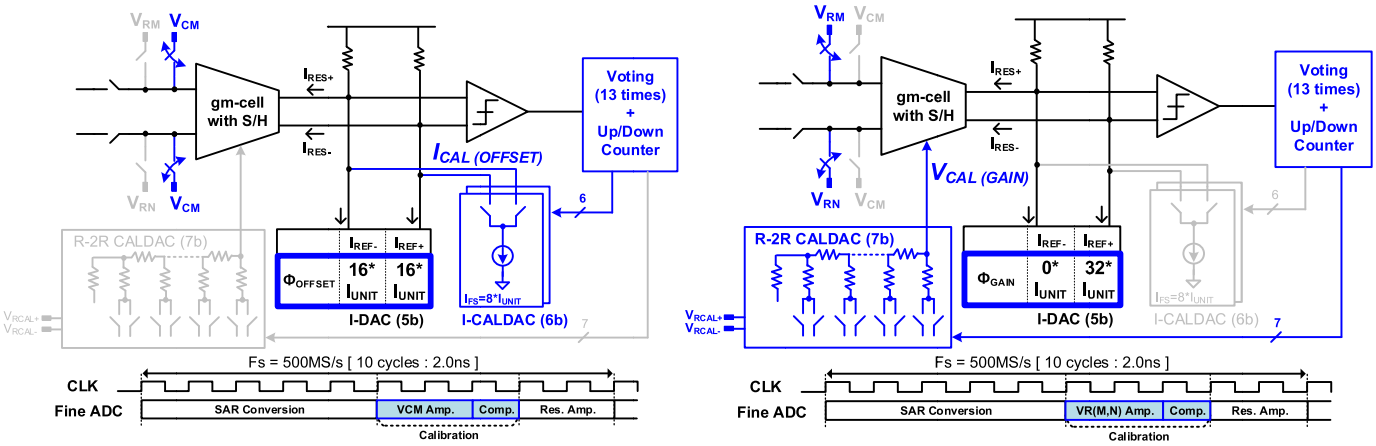
As shown in Fig. 5, owing to the small R_T, all the current sources in the current-steering DAC are designed with sufficient voltage headroom for high linearity even under a 1.0-V supply. The unit current cell of the 400-μA-full-scale 5-bit I-DAC consists of a current source (M_{CS}), a cascode transistor (M_{CAS}), and a switching transistor (M_{SW}, M_{SWB}). The size of the unit current source (M_{CS}) is designed with a sufficient margin to achieve a relative standard deviation of $<2.7\%$, corresponding to about an 8-bit static linearity [18]. With the given voltage swing level at the output of the DAC, the stacked transistors in the saturation region (M_{CAS}, M_{SW}, and M_{SWB}) achieve 100 times larger output resistance than the required value of 160 kΩ [19]. The trans-conductance of the unit current source (M_{CS}) is designed to be as low as possible for the noise performance of the fine ADC. With small switch transistors (M_{SW}, M_{SWB}) and a compact layout of the DAC [20], this work minimizes the parasitic capacitance (C_P) for high-speed residue amplification and high-speed SAR operation.

C. Speed and Noise Analysis

The dominant pole of $1/(R_T \cdot C_P)$ is the common factor that determines the settling time of the gm-cell in the sampling phase and that of the I-DAC for SAR conversion. As shown in Fig. 3, the time of one cycle of the synchronous clock is 200 ps. As the comparator latching and digital logic in the 5-bit fine SAR ADC take about 150 ps in each settling, 50 ps can be assigned for I-DAC settling. Thus, with R_T of 320 Ω, C_P must be smaller than 45 fF. The designed (estimated) parasitic capacitance (C_P) in Fig. 5 is about 30 fF, consisting of 15-fF routing capacitance and 15-fF device parasitic including 9 fF by the 5-bit I-DAC. The resultant small time-constant ($R_T \cdot C_P$) of 9.6 ps also enables fast residue settling of about 250 ps even with the internal parasitic of the gm-cell with S/H, making two clock cycles assigned for residue amplification to be reasonable.

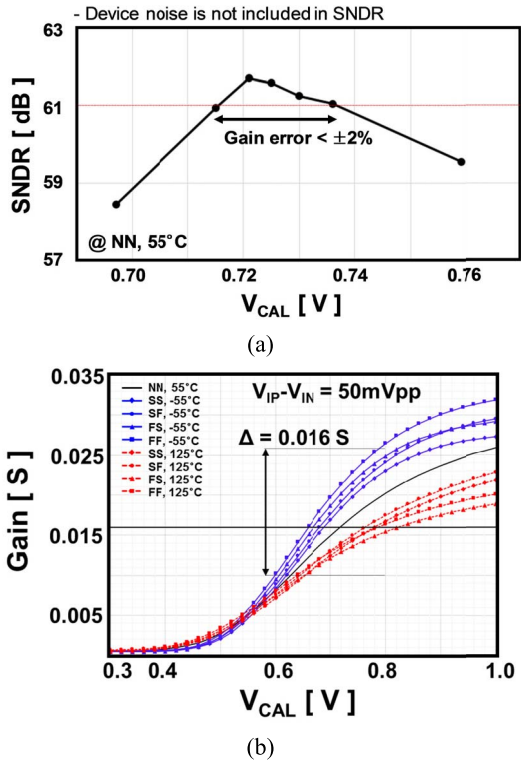
Fig. 12 shows the noise analysis result of the gm-cell with S/H. The noise of the gm-cell is composed of those from the FVF, the sampling network (switch and C_G) in sampling mode (Φ_{IS} = H) and additional noise from the V–I and its output parasitic capacitance (C_P) in holding mode ($\Phi_{IS} = L$). The simulated input-referred noise of the gm-cell with S/H is 180.67 nV^2, mainly caused by the noise of the FVF and the sampling switch. Although a large value of C_G could

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(a) (b)

Fig. 13. Background calibration (a) in offset and (b) in gain.



(a)

(b)

Fig. 14. Simulation results of (a) ADC's signal-to-noise-and-distortion ratio (SNDR) versus V_{CAL} in the gm-cell and (b) gain versus V_{CAL} in the gm-cell.

improve the noise performance of the gm-cell, it degrades the operating speed. The C_G, 15 fF, is implemented with only gate capacitance of the mirror transistor (M5, M6 in Fig. 5) to maximize the speed while the noise of the gm-cell is smaller than the quantization noise in ADC's SNR. Table I shows the simulated noise contribution of the overall ADC. Owing to the noise optimization in the fine ADC, the noise contribution of the fine ADC even with small parasitic capacitance (C_P) is only 10.1% of the total. Hence, the proposed ADC could be designed to achieve an SNR higher than 55 dB with no band-limiting additional capacitors for high speed.

TABLE II
SIMULATED OFFSET VALUE

Current-mode Back-end	Current output offset	Input-referred offset
gm-cell	30.6 μA	1.22 LSB
gm-cell + I-DAC	32.1 μA	1.28 LSB
gm-cell + I-DAC + Comp.	38.4 μA	1.54 LSB

D. Background Calibration

With open-loop residue amplification, as is well known, inter-stage errors should be addressed. In this architecture, the gain mismatch between the full-scale current of the gm-cell and that of the fine current-steering DAC should be calibrated. In this work, a zero input and the positive residue full-scale are taken as two reference points for two-point gain calibration, which inherently corrects the offset error. The proposed two-point gain calibration scheme is conducted in the background with an on-chip calibration block. Both the gain and offset calibration operations share the same clock periods in an alternating manner, as illustrated in the timing diagrams in Fig. 13(a) and (b).

As shown in Fig. 13(a), the offset error is calibrated by applying a differential zero input to the gm-cell after the fine A/D conversion is completed. During this phase, the 5-bit I-DAC in the fine SAR ADC is reset for zero output. The comparator output then indicates the polarity of the combined offset of the gm-cell and I-DAC. By adding a current-mode calibration DAC (I-CALDAC) at the output of the I-DAC, the offset is adjusted by the offset canceling current, I_{CAL}, by referring to the comparator output. Table II shows the Monte-Carlo simulated offset values of the gm-cell and fine ADC. The 1 sigma value (σ) of the total current offset of the current-mode back-end circuit (before calibration) is 38.4 μA, dominated by the effects of the gm-cell and the comparator. Based on the simulated input-referred offset of a 1.54 LSB, a 6-bit I-CALDAC is designed with a differential full-scale current of 200 μA (coverage of 5.2 sigmas), which can adjust the input-referred offset with a step size of 0.125 LSB.

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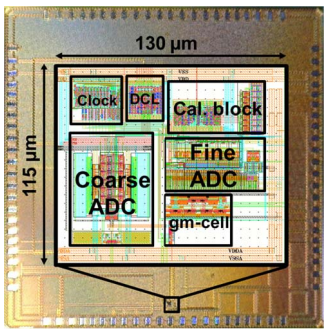


Fig. 15. Die photograph and layout.

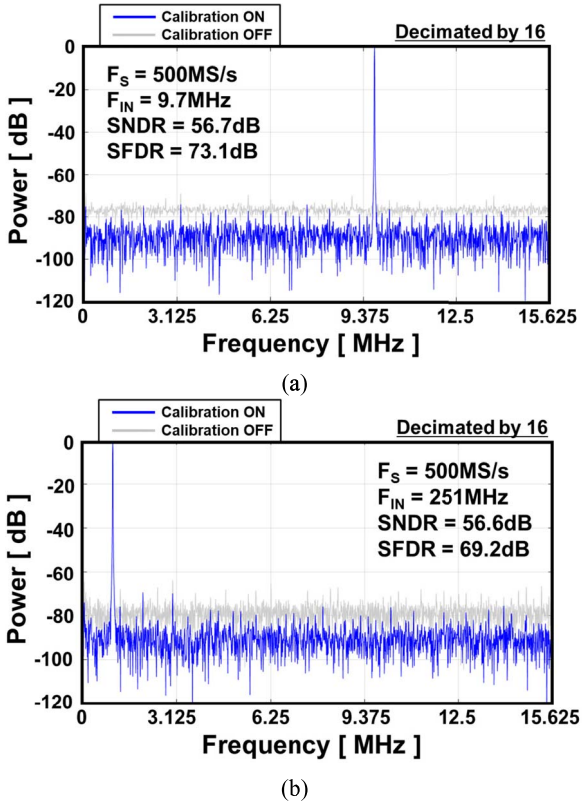


Fig. 16. Measured FFT spectrum at 500 MS/s with (a) 9.7- and (b) 251-MHz input.

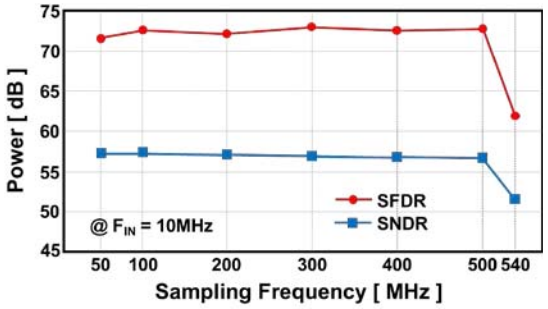


Fig. 17. Measured SNDR and spurious-free dynamic range (SFDR) versus sampling frequency.

For gain error calibration, as shown in Fig. 13(b), the gm-cell is connected to the reference voltages that represent the full-scale residue, V_{RM} and V_{RN}, and the 5-bit I-DAC is set

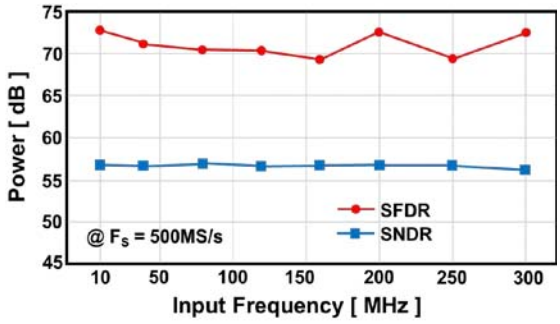


Fig. 18. Measured SNDR and SFDR versus input frequency.

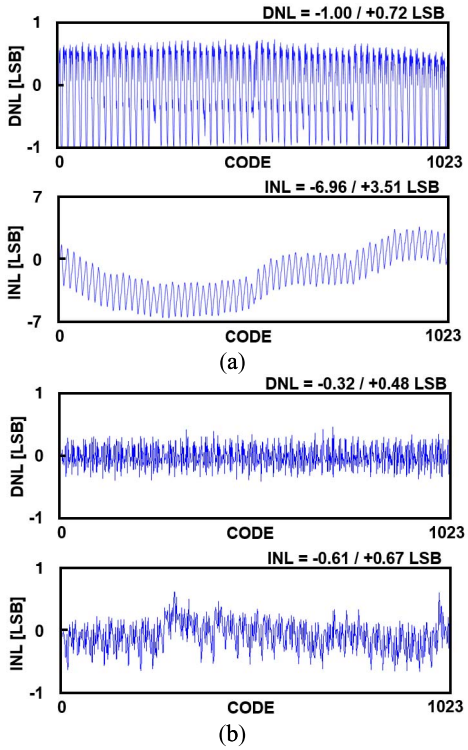


Fig. 19. Measured differential non-linearity (DNL) and integral non-linearity (INL) (a) before calibration and (b) after calibration.

to its full-scale value ($32*I_{UNIT}$). An additional calibration DAC (7-bit R-2R CALDAC) adjusts the degeneration resistor value (MR1 in Fig. 5) of the gm-cell with V_{CAL} so that the output current from the gm-cell converges to the full-scale current of the I-DAC. For an SNDR (without device noise) over 61 dB, as simulated in Fig. 14(a), the allowed gain error range is less than $\pm 2\%$ of the full-scale current of the I-DAC. Fig. 14(b) shows the voltage-to-current conversion gain change according to the V_{CAL} in the various process corners and temperature conditions. The worst-case simulated value of the gain variation is about 0.016 S at the edge of V_{CAL} of 0.7 V. Hence, the 7-bit R-2R CALDAC has a control range of about 200 mV and a step size below 1% for gain control. The two references, V_{RM} and V_{RN}, are generated utilizing a resistive divider from an additional ADC reference, considering a residue gain drop of about 20% due to parasitic

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TABLE III
PERFORMANCE SUMMARY AND COMPARISON

	VLSI16 Y.-C. Lien [8]	VLSI13 B. Verbruggen [6]	TVLSI17 Y. Zhu [7]	JSSC17 H. Huang [9]	ISSCC18 L. Kull [10]	This Work
Technology	28 nm	28 nm	65 nm	65 nm	14 nm	28 nm
Resolution [bit]	12	11	10	12	10	10
Residue amp. Structure	Gain-boosting telescopic amplifier	Dynamic amplifier	Gain-boosting telescopic amplifier	Dynamic amplifier	CS amplifier	gm-cell
F_S [MS/s]	800	410	500	330	1500	500
# of TI channels	4	2	2	-	-	-
F_S per channel [MS/s]	200	205	250	330	1500	500
Supply Voltage [V]	1.0	1.0	1.2	1.2	0.95	1.0
SNDR @Nyquist [dB]	60.8	55	55.9	63.5	50.1	56.6
Power [mW]	14.6	2.1	8.2	6.23	6.92	6.0
FoM_{Walden} @Nyq. [fJ/conversion-step]	20.4	11.4	32.2	15.4	17.7	21.7
$FoM_{Schreier}$ @Nyq. [dB]	165.2	164.9	160.7	167.8	160.5	162.8
Area [mm^2]	0.093	0.11	0.046	0.08	0.0016	0.015

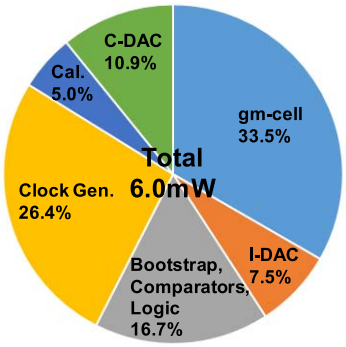


Fig. 20. Power consumption breakdown.

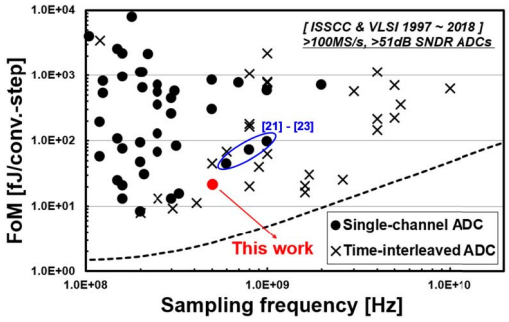


Fig. 21. FoM comparison with published ADCs [24].

capacitance at the top-plate node of the coarse ADC. Thirteen times of voting are conducted with the comparator decisions for a single decision of error polarity for improved calibration accuracy. Note that the required input-reset of the gm-cell after calibration is actually conducted during the "Comp" period in Fig. 13(a) and (b) as the FVF is disconnected from the V–I in hold mode.

IV. MEASUREMENT RESULTS

A prototype pipelined-SAR ADC with the current-mode residue processing was implemented in a 28-nm CMOS process. Fig. 15 shows the chip photograph with an active area of 0.015 mm^2. Additional hardware involved in the calibration for inter-stage mismatch errors accounts for approximately 10% of the total area of the ADC, as shown in the layout. All the measured results are achieved at a 500-MS/s conversion rate and a 1.0-V power supply. Fig. 16(a) and (b) show the measured fast Fourier transform (FFT) spectrums decimated by a factor of 16 with and without background calibration. When calibrations for both offset and gain error are enabled, at a 9.7-MHz input, the SNDR and SFDR of the ADC are 56.7 and 73.1 dB, respectively. With an input at 251 MHz, the measured SNDR and SFDR are 56.6 and 69.2 dB, respectively. Fig. 17 shows the ADC's dynamic performance for various sampling rates with a 10-MHz input, presenting an SFDR higher than 72 dB up to a 500-MS/s conversion rate. Fig. 18 shows the dynamic performance for various input frequencies at a 500-MS/s sampling rate. It shows quite a constant performance, maintaining an SNDR above 56 dB up to the input frequency of 300 MHz. Fig. 19 shows the linearity performances before and after calibration. Before calibration, the DNL shows many missing codes. After calibration is enabled, the DNL and INL are −0.32/+0.48 and −0.61/+0.67 LSB, respectively. As the power breakdown presented in Fig. 20, the total power consumption is 6 mW, including 2 and 1.6 mW by the gm-cell and the clock buffer, respectively.

Table III summarizes the performance of the prototype ADC and compares it with previously reported high-speed pipelined SAR ADCs. This prototype ADC achieves the highest single-channel conversion-rate among ADCs with various residue amplification techniques, except for the case of the FinFET process [10]. These results show that the proposed ADC with a current-mode residue processing scheme is a promising pipelined-SAR architecture for high-speed

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applications. Fig. 21 compares the figure of merit (FoM) of this work with previously reported ADCs. Note that this work achieves low power consumption with a faster residue processing period (<400 ps) relative to other high-speed pipeline ADCs [21]–[23], showing a remarkable FoM of 21.7-fJ/conversion-step among the published single-channel ADCs, which are indicated with black dots.

V. Conclusion

This paper introduces a current-mode open-loop residue processing technique with a SAR-based pipelined architecture. The residue processing in the current domain enhances the operation speed in the pipelined SAR ADC, owing to the low-impedance signaling and the high-speed split-current source DAC switching. An area-efficient background calibration function is embedded on a chip to resolve the inter-stage mismatch issues in the open-loop voltage-to-current conversion. The single-channel 10-bit 500-MS/s prototype ADC achieving an FoM of 21.7-fJ/conversion-step successfully verified the design concept.

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Kyoung-Jun Moon received the B.S. and M.S. degrees in electronic engineering from Sogang University, Seoul, South Korea, in 2005 and 2007, respectively. He is currently pursuing the Ph.D. degree in electrical engineering with the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.

He joined Samsung Electronics, Hwaseong, South Korea, in 2007, and has developed data converter (ADC/DAC) circuits. His research interests include high-speed low-power data converter and mixed-signal circuits.



Dong-Shin Jo received the B.S. degree in electrical engineering from Kyungpook National University, Daegu, South Korea, in 2006, and the M.S. and Ph.D. degrees from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2012 and 2019, respectively.

He is currently with SK Hynix, Icheon, South Korea, where he designs a high-speed memory interface. His research interests include power-efficient high-speed ADC and low-power ADC for bio-applications.



Wan Kim received the B.S. degree (Hons.) in electronic engineering from Inha University, Incheon, South Korea, in 2008, the M.S. degree in information and communications from the Gwangju Institute of Science and Technology, Gwangju, South Korea, in 2010, and the Ph.D. degree in electrical engineering from the Korea Advanced Institute of Science and Technology, Daejeon, South Korea, in 2016.

Since 2016, he has been a Senior Engineer with Samsung Electronics, Hwaseong, South Korea. His current research interests include digital calibration techniques for high-resolution data converters and low-noise mixed-signal circuit design for low-power applications.

Dr. Kim was a recipient of a silver prize in the Samsung HumanTech Paper Award in 2016.

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Michael Choi received the master's and Ph.D. degrees from the University of California at Los Angeles, Los Angeles, CA, USA, in 1998 and 2002, respectively.

He joined Samsung Electronics, Hwaseong, South Korea, in 2006, where he is currently a Master. His expertises include high-speed data converters and various analog front ends for UHD digital TV, WiFi and 5G connectivity, automotive V2X, touch controller, and CMOS image sensor.



Hyung-Jong Ko received the B.S. degree in electronic engineering from Ajou University, Suwon, South Korea, in 1995, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Florida, Gainesville, FL, USA, in 1998 and 2002, respectively.

Since 2003, he has been with Samsung Electronics Corporation, Hwaseong, South Korea, where he is currently the Vice-President, focusing on the design of mixed-signal circuits and responsible for the development of analog IPs for various products.



Seung-Tak Ryu (M'06–SM'13) received the B.S. degree in electrical engineering from Kyungpook National University, Daegu, South Korea, in 1997, and the M.S. and Ph.D. degrees from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 1999 and 2004, respectively.

From 2001 to 2002, he was a Visiting Researcher with the University of California at San Diego, San Diego, CA, USA, sponsored through the Brain Korea 21 (BK21) Program. In 2004, he joined Samsung Electronics, Giheung, South Korea, where he was involved in the mixed-signal IP design. From 2007 to 2009, he was an Assistant Professor with the Information and Communications University (ICU), Daejeon. He has been with the Department of Electrical Engineering, KAIST, since 2009, where he is currently an Associate Professor. His current research interests include analog and mixed-signal IC design with an emphasis on data converters.

Dr. Ryu served as a member of the Technical Program Committee (TPC) for the International Solid-State Circuits Conference (ISSCC), and he is currently a TPC Member of the Asian Solid-State Circuits Conference (A-SSCC) and the Custom Integrated Circuits Conference (CICC). He has also served as a Guest Editor for the IEEE JOURNAL OF SOLID-STATE CIRCUITS and is currently serving as an Associate Editor for the IEEE SOLID-STATE CIRCUITS LETTERS.

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